NLP Climate Project

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1 Introduction

In this project, we apply text classification techniques to analyze how sentiment toward climate change has evolved on Twitter over time. Specifically, we investigate the question: How has climate sentiment on tweets changed across key political and environmental milestones, such as the 2015 Paris Climate Accords? Using BERT-based models and sentiment analysis methods covered in class, we aim to classify and track sentiment trends across multiple years. This problem is not only technically relevant to natural language processing and machine learning, but also socially significant, providing insights into how public opinion responds to major global events and policy shifts. Our novel contribution lies in combining temporal analysis with sentiment classification to reveal changes in climate discourse on social media, with potential extensions into semantic drift and public policy evaluation.

2 Literature Review

The time periods between 27 April 2015 and 21 February 2018, as well as between 2021 and 2023, are two crucial moments in history tied to world events such as the signing of the climate accords in Paris on 12 December 2015, and the emergence of the pandemic in 2020. These two periods are ideal for narrowing in on how climate change sentiment is affected by major world events. Many sources have addressed both climate change sentiment as well as sentiment on other topics that are useful for our research.

The first that we will cover is another paper that specifically used VADER sentiment analysis, in this case to perform a sentiment analysis of tweets on the topic of bitcoin titled "A complete VADER-based sentiment analysis of bitcoin (BTC) tweets during the era of COVID-19" (Pano and Kashef, 2020). This paper is also of interest as the sentiment analysis performed is during the COVID pandemic, which is a similar period to our own. Some notable features of this paper include the detection of emoticons such as ":)" and ";(" to recognize positive or negative sentiment, as these characters are often used on Twitter and about this topic. This is a feature that we will likely include. The graphs of VADER polarity included in Bitcoin can also be a good metric to compare with our own, and specifically, the graph of the average VADER sentiment per minute may be especially helpful when constructing visualizations. (Pano and Kashef, 2020)

Another paper that may be quite useful is titled "Sensing climate change and energy issues: Sentiment and emotion analysis with social media in the UK and Spain" (Loureiro and All , 2020)). Like the title suggests, this is a similar sentiment analysis to our own paper, but with a specific focus on the geographical locations of the United Kingdom and Spain. In this paper, they also analyze Tweets, they include some great figures, such as social media activity (number of daily tweets) in each country, and an analysis of Google search trends that include climate change. Also included is sentiment analysis towards specific types of climate change related topics such as coal, nuclear energy, etc. These could be interesting to include in our paper and to analyze sentiments about climate change and relevant topics as well. We also see word clouds, which are great for visualization and which we will include in our paper (Loureiro and All , 2020).

Moving on to the importance of visuals, the next paper contains some of the best graphs and imagery we have found in any on this topic. The title reads "Exploring climate change on Twitter using seven aspects: Stance, sentiment, aggressiveness, temperature, gender, topics, and disasters" (Effrosynidis et al., 2022), and from the title alone, it is clear that this paper is quite relevant to our research. Specifically, this paper focuses heavily on these specific

aspects.

The concept of universal language model fine-tuning, introduced by Howard and Ruder (2018), further also supports our approach. Their ULMFiT framework demonstrated that even with limited labeled data, language models pretrained on general corpora can be effectively adapted to specific classification tasks through techniques like discriminative fine-tuning, slanted triangular learning rates, and gradual unfreezing. These strategies are crucial in our scenario, where class imbalance may hinder training stability. By adopting these techniques with BERT, we aim to mitigate overfitting to the dominant class and improve generalization.

Next, A valuable comparison between dictionary-based and transformer-based sentiment models is offered by Feldkamp et al. (2024), who evaluated tools like VADER and RoBERTa across different literary texts. They found that VADER performed competitively on short, stylized texts—like tweets—while RoBERTa captured more nuanced sentiment arcs. These findings validate our plan to compare VADER (lexicon-based, rule-driven) with BERT (contextual, deep learning) in classifying climate-related tweets. This dual-model approach is critical not only for performance benchmarking but also for investigating whether general sentiment (e.g., positive/negative) aligns with issue-specific sentiment (e.g., pro/anti-climate).

In Atefinia and Mirnezami (2025)'s examination of Iranian discourse on the Paris Agreement offers insights that align with our own climate-sentiment study. The authors compare sentiment volume with an event timeline, demonstrating that negative peaks coincide with exogenous shocks—Trump's 2017 withdrawal and the 2021 Iranian blackout crisis—illustrating the value of overlaying sentiment analysis with real world events. Reproducing their pipeline on our data could test whether analytic choices, not just cultural context, shape observed distributions. Complementing the qualitative themes, the study also looked at things like word counts and hashtags. We can adopt this approach and include similar analysis, such as per-class word counts visualized as word clouds.

Agarwal et al. (2011) discusses some general best practice for gathering and analyzing twitter data for sentiment. They explore a tree kernel approach that minimizes the need for extensive feature engineering while still achieving competitive performance compared to traditional models. This could be a potential point of experimentation to improve our model. While this paper came before transformer models like BERT, it is still another interesting model to compare. While we already have a dataset to train on, this paper could help us if we want to gather current data to see what the sentiment is today.

Finally, an interesting analysis we could try is misinformation detection. Our dataset comes with a "news" tag, which we could use to train a model to classify tweets containing misinformation. In Hossain et al. (2020), they analyze COVID misinformation on social media. Their methods could prove very useful, even outside of misinformation detection. They used models like Sentence-BERT (SBERT) (Reimers and Gurevych, 2019), which we talked about in class and could be better for tweet classification than standard BERT. SBERT is a modification of BERT that uses Siamese and triplet network architectures to produce fixed-size, semantically meaningful sentence embeddings by fine-tuning on natural language inference data. These embeddings make it efficient to compare or cluster tweets (e.g. via cosine similarity) and feed them into lightweight classifiers. Its efficiency not only helps us train faster but also has less energy consumption and environmental effects.

3 Data

We use two datasets to train, test, and apply sentiment classification models on climate-related tweets across different time periods.

3.1 Training and Testing Data

The first dataset is titled "Twitter Climate Change Sentiment Dataset" and includes 43,943 climate-related tweets collected between April 27, 2015, and February 21, 2018. Each tweet was independently labeled by three human annotators, and only those tweets with full agreement among all annotators were retained. Tweets are labeled according to their stance on climate change:

- **1 (Pro):** Supports the belief in man-made climate change
- **0 (Neutral):** Neither supports nor refutes man-made climate change
- **-1 (nti):** Denies the belief in man-made climate change

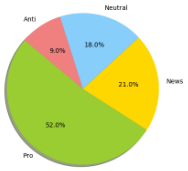


Figure 1: Distribution of Data by Classes

This dataset serves as the foundation for training and evaluating our models. Below is a breakdown of the distribution of the labeled data.

3.2 Prediction Dataset

The second dataset is titled "Tweets About Climate Change and Joe Biden" and consists of 5467 unique values of climate change-related tweets collected between 2021-2023 using the Tweepy API. Unlike the first, this dataset is unlabeled and will be used to predict sentiment using the trained model. This second dataset is particularly relevant for assessing public opinion on recent climate policies, including those introduced by the Biden administration. By applying our model to this dataset, we aim to track sentiment trends, identify prevailing concerns, and evaluate how political developments shape public discourse around climate change.

4 Methods

This project implements and compares two sentiment classification approaches: a BERT-based deep learning model and a rule-based VADER baseline to evaluate how public sentiment toward climate change has evolved on Twitter.

4.1 Data Preparation and Preprocessing

To ensure consistent input for both models, we applied a standardized preprocessing pipeline:

- Removal of retweet headers (`RT @user:`) and user mentions (`@user`)
- Stripping of URLs and hyperlinks
- Conversion of all text to lowercase
- Removal of tweets labeled as "news" (label 2), retaining only -1 (anti), 0 (neutral), and 1 (pro)

This preprocessing was implemented in a single `clean_message )` function applied across the dataset, with code maintained in version-controlled scripts. Stratified sampling was used to split the data to guarantee balanced representation across classes and improve generalizability.

The cleaned dataset was split as follows:

- Training set (64%)
- Development (validation) set (16%)
- Test set (20%)

This two-step splitting process ensures a proper development workflow. First, 20% of the data was held out as the test set. The remaining 80% was further split into training and development sets using an 80/20 ratio. We used a fixed random seed (`random_state=42`) to ensure full reproducibility. Each split preserves the original class distribution through stratified sampling to prevent sampling bias. All processed datasets were saved as CSV files to ensure reproducibility and enable external validation.

4.2 Baseline Model: V DER

The **V DER** (Valence Aware Dictionary and sEntiment Reasoner) model from the NLTK library was used as a non-trainable, rule-based sentiment analyzer. Each tweet's sentiment was determined based on its compound polarity score:

- $>$: *Pro* (belief in man-made climate change)
- $=$: *Neutral*
- : *nti* (denial of man-made climate change)

Predictions were mapped to the same three-class format as the labeled dataset and evaluated using precision, recall, and F1-score. VADER serves as a reproducible and interpretable benchmark for evaluating the performance of our fine-tuned deep learning model.

4.3 BERT model

We fine-tuned a pre-trained **DistilBERT** model (`distilbert-base-uncased`) using the Hugging Face Transformers and PyTorch libraries. The model architecture includes:

- The base DistilBERT transformer
- A dropout layer (0.1) to prevent overfitting
- A final linear layer projecting to 3 sentiment classes

Tokenization: Tweets were tokenized using DistilBERT's tokenizer with truncation, padding, and a maximum sequence length of 128.

Label Encoding: For compatibility with PyTorch's `CrossEntropyLoss`, labels were remapped from {-1, 0, 1} to {0, 1, 2}.

Training Setup:

- Loss Function: `CrossEntropyLoss`
- Optimizer: `AdamW`
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score
- Device: CPU/GPU autodetection
- Training Framework: Hugging Face `Trainer` with structured logging and checkpointing

Hyper-parameters:

- Learning rate: `1.827e-05`
- Batch size: `8`
- Epochs: `5`
- Weight decay: `0.168`
- Warmup steps ratio: `0.118`
- Max length: `128`
- Early stopping patience: `5`

4.4 Naive Bayes Model

We also implemented a traditional Naive Bayes classifier using bigram and trigram features. The model pipeline includes:

- Lowercasing and basic text cleaning
- Extraction of n-gram features (2-grams and 3-grams)
- Training on the labeled dataset with class labels remapped as needed

The model was serialized using `joblib` to ensure reusability, and evaluations were conducted on both the development and test sets using accuracy and classification reports.

4.5 Evaluation and Reproducibility

To ensure methodological soundness, all three models were evaluated on the same test set using identical preprocessing, label formatting, and evaluation metrics. Classification reports were generated to provide detailed performance breakdowns per sentiment class.

All experiments were implemented in modular scripts with fixed random seeds and documented configurations (e.g., training arguments, label mappings). Intermediate results and final outputs (e.g., predictions, models, evaluation metrics) were saved in structured directories for reproducibility.

4.6 Sentiment Prediction on Unlabeled Data

After training, the BERT model was applied to a second dataset of unlabeled climate-related tweets. This demonstrated the model's use in real-world inference, allowing exploration of sentiment trends over time and in relation to policy shifts. The dataset also includes metadata such as location and topics, supporting potential future extensions such as geospatial or topic-based analysis.

5 Evaluation

To assess the performance of both models, we evaluated them on the same held-out test set using standard classification metrics: precision, recall, F1-score, and overall accuracy. Below are the classification reports for both VADER baseline and the BERT-based model .

5.1 V DER Baseline Results

The VADER model performed significantly worse, with an overall accuracy of 35%. While VADER showed moderate recall on the *anti* class, it struggled to capture nuanced distinctions between sentiment categories, especially for *neutral* tweets. This illustrates the limitations of rule-based approaches in handling complex or context-dependent sentiment.

Class	Precision	Recall	F1-score	Support
Pro	0.61	0.36	0.45	4593
Neutral	0.25	0.24	0.25	1543
Anti	0.14	0.47	0.21	798

Table 1: Classification report for VADER sentiment classifier

5.2 Naive Bayes Classification Results

The Naive Bayes model showed strong performance on the *pro* class, achieving an F1-score of 0.85 and high recall of 0.96. However, performance on the *neutral* and *anti* classes was weaker, with F1-scores of 0.46 and 0.47 respectively. This is because of the data imbalance in out dataset. The overall accuracy was 75.0%.

Class	Precision	Recall	F1-score	Support
Pro	0.76	0.96	0.85	4593
Neutral	0.67	0.35	0.46	1543
Anti	0.79	0.34	0.47	798

Table 2: Classification report for Naive Bayes sentiment classifier

5.3 BERT Classification Results

The BERT model achieved strong overall performance, particularly on the *pro* class, with an F1-score of 0.883. While performance was lower on the *neutral* and *anti* classes, the macro-average F1-score of 0.726 and an overall accuracy of 80.4% demonstrate the model's effectiveness in multi-class sentiment classification.

Class	Precision	Recall	F1-score	Support
Pro	0.863	0.904	0.883	4593
Neutral	0.642	0.600	0.620	1543
Anti	0.734	0.627	0.676	798

Table 3: Classification report for BERT-based sentiment classifier

5.4 Summary

The BERT model substantially outperformed both VADER and Naive Bayes across all major metrics. This highlights the strength of transformer-based models in capturing nuanced and context-dependent sentiment, particularly in complex and evolving domains like climate change discourse.

While Naive Bayes performed better than VADER and showed high recall on the *pro* class, it struggled to accurately classify *neutral* and *anti* tweets. This suggests that although traditional machine learning methods can be useful with n-gram features, they lack the deep semantic understanding provided by pre-trained language models.

Overall, the results underscore the limitations of rule-based approaches like VADER in domain-specific sentiment analysis and support the adoption of data-driven, contextual models like BERT for high-stakes applications such as tracking public opinion on climate policy.

5.5 Results: Sentiment Predictions on Biden-Era Climate Tweets

To explore contemporary sentiment around climate change, we applied each of the three trained models, BERT, Naive Bayes, and VADER, to an unlabeled dataset of tweets collected during the Biden administration. The results reveal notable variation in how each model interprets sentiment:

- **BERT** predicted a majority of tweets as *pro* (59.9%), followed by *neutral* (23.7%) and *anti* (16.3%). This suggests a generally optimistic public discourse around climate change under the Biden administration.
- **Naive Bayes** classified most tweets as *neutral* (94.1%), with only 3.7% labeled as *anti* and 2.2% as *pro*. This indicates a strong bias toward neutrality, possibly due to the model's reliance on sparse n-gram features, which may limit its sensitivity to more polarized expressions.
- **V DER** returned the most pessimistic outlook, labeling 43.1% of tweets as *anti*, 36.8% as *pro*, and 20.0% as *neutral*. This pattern may reflect VADER's lexicon-based structure, which can struggle to accurately interpret nuanced or sarcastic content.

Model	Pro (%)	Neutral (%)	nti (%)
BERT	59.9	23.7	16.3
Naive Bayes	2.2	94.1	3.7
VADER	36.8	20.0	43.1

Table 4: Sentiment predictions by each model on Biden-era climate tweets

- Majority-Class Baseline Comparison

Also important to mention is our comparisons with the majority-class baseline. Our comparisons indicated that even Naive Bayes performed better than the Majority Baseline. Majority Class was at 60 percent accuracy while Naive Bayes had 75 percent accuracy. This demonstrates that the Naive Bayes, and of course BERT, significantly outperformed this baseline in our accuracy measurements.

These differing outputs illustrate how model architecture and methodology significantly influence

sentiment interpretation, especially in domains nuanced and politically sensitive as climate d course.

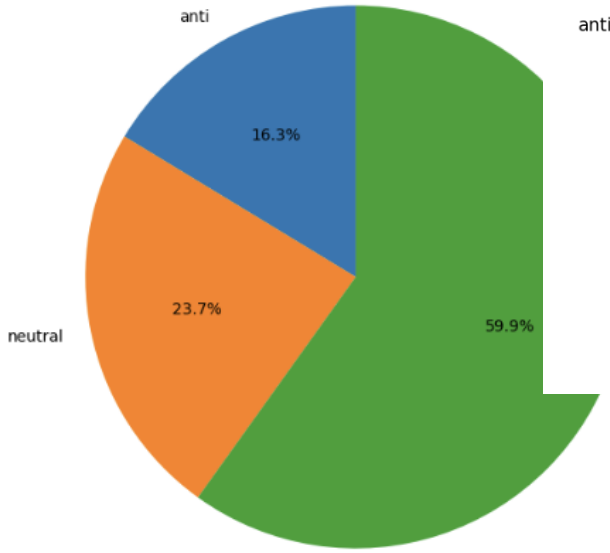


Figure 2: BERT-predicted sentiment distribution on Biden-era tweets

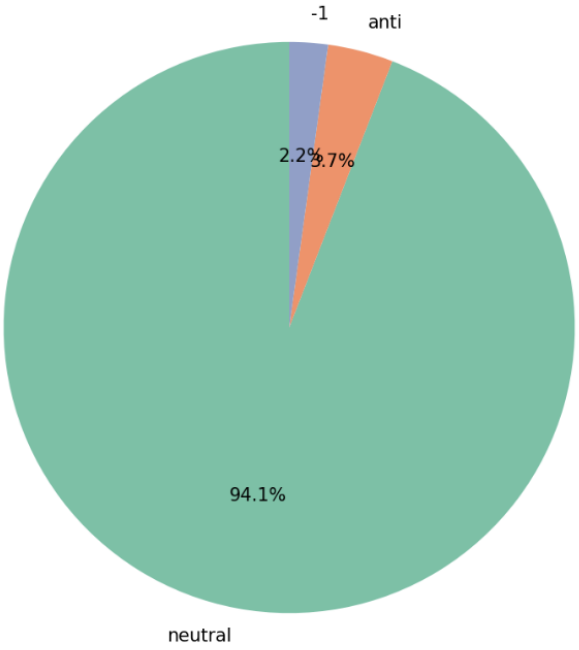


Figure 3: Naive Bayes predicted sentiment distribution on Biden-era tweets

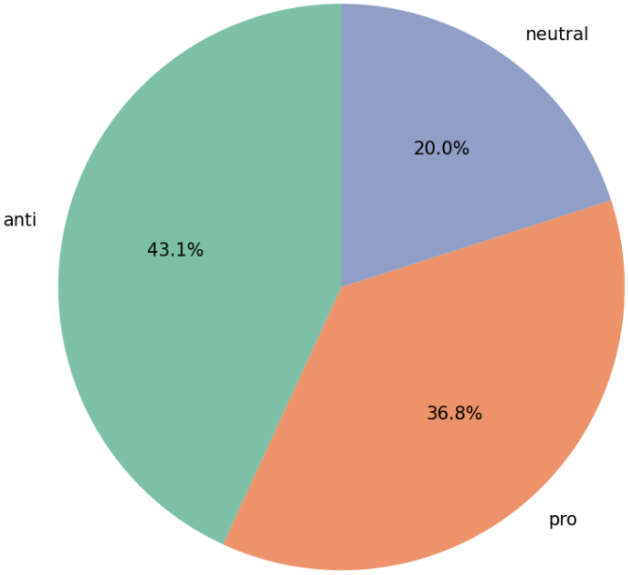


Figure 4: VADER-predicted sentiment distribution on Biden-era tweets

5.6 Error nalysis: Interpreting Sentiment in Light of Biden's Climate Policies

The Biden administration marked a sharp shift toward climate-forward policies, rejoining the Paris Climate Agreement on the first day in office, passing the Inflation Reduction Act with historic clean energy investments, and promoting climate justice and international cooperation. These policy moves provide a meaningful backdrop for interpreting the sentiment distributions observed in our models' predictions.

- **BERT**'s prediction of a majority *pro* sentiment (59.9%) aligns with the administration's climate agenda and suggests that public discourse on Twitter during this period was generally supportive of environmental reforms. The relatively low percentage of *anti* sentiment (16.3%) further supports this interpretation, indicating positive reception of climate action among a significant portion of Twitter users. This, aligned with the even more positive sentiment on the training data may suggest that pulling out of the climate accords did lead to less positive sentiment, but still overall a pro aligned. This might make sense in the general context of the mentioned Biden administration actions.
- **Naive Bayes** classified most tweets as *neutral* (94.1%), likely reflecting its limited capacity to interpret nuanced or indirect sen-

timent. This may also suggest that many climate-related tweets during this period were informational or policy-focused rather than overtly opinionated, which the model interpreted as neutral. It highlights the need for more context-aware approaches when analyzing political discourse.

- **V DER** produced a more pessimistic view, labeling 43.1% of tweets as *anti*. This may stem from the lexicon-based nature of VADER, which struggles with contextual cues such as sarcasm or complex critiques. It may also reflect ongoing polarization and skepticism surrounding government climate initiatives, particularly from political opposition or climate denialist voices.

The divergence between model predictions underscores the critical role model architecture plays in interpreting public sentiment. Transformer-based models like BERT offer a more reliable lens for analyzing evolving, domain-specific discourse due to their contextual understanding. In contrast, rule-based or traditional statistical models risk oversimplifying or misclassifying sentiment in politically charged environments. These findings highlight the importance of selecting appropriate NLP tools for sentiment analysis in sociopolitical contexts such as climate change.

6 Ethical Considerations

All tweets in both the training and prediction datasets are publicly available posts that were collected in accordance with the Twitter Developer Policy. No attempts were made to deanonymize authors. We only use raw messages, no user data in our training. We acknowledge that public availability does not equate to informed consent; future work should explore mechanisms for honoring users' data-sharing preferences (e.g. opt-out lists).

Twitter's user base is younger, more urban, and more politically engaged than the general population. Consequently, our sentiment estimates may not generalize to offline discourse. In addition, the labeled corpus is highly imbalanced (*pro*>*neutral*>*anti*), which can lead to inflated performance in the majority classes and systematic under-detection of minority viewpoints. We tried to mitigate these issues with sampling, class-weighted loss functions, and per-class reporting, but residual bias is likely. Interpretation of the results should therefore emphasize *relative* changes over time rather than absolute population-level sentiment.

The 2015-2018 dataset was annotated by crowd workers. Although we retained only items with unanimous agreement, annotator demographics and working conditions were not recorded, so we cannot fully assess positional or cultural bias. We urge caution when deploying the model for high-stakes applications and recommend periodic audits with domain experts.

Large language models are resource-intensive. We opted for `distilbert-base-uncased`—a 40% smaller, 60% faster variant of BERT—to reduce energy consumption while maintaining performance.

7 Conclusions

The central takeaways from our research fall into a couple of categories, as we can take away information both on the accuracy of the chosen models we implemented, but also on general trends in sentiment on the political-social topic of climate change and the public's reaction to global events. We can see from the data that between the 2015-2018 data and the 2023 Data there is a clear uptick in negative climate sentiment by 4.8 percent and a clear decrease in positive sentiment by 6.3 percent. This may answer some of our initial questions about the change in sentiment between these two periods, especially as these two periods are heavily defined politically by different presidencies as well as the Paris Climate Accords and the trend of worsening of climate change. Although it would be a step to apply any type of direct causal relationship, these data collected are an interesting change in sentiment surrounded by the events discussed in the introduction and literature review, and so could be cited in a more thorough analysis on the effects of the climate accords and world events on public perception and attitude. The other useful conclusions that come from our research include more concrete information on the most appropriate approach for this sentiment analysis between the VADER, BERT, and Naive Bayes models. As described, BERT vastly outperformed the other two, which demonstrates the importance of these contextual models among sentiment analysis use cases with access to significant data.

References

Apoorv Agarwal, Boyi Xie, Ilia Vovsha, Owen Rambow, and Rebecca Passonneau. 2011. Sentiment analysis

of Twitter data. In *Proceedings of the Workshop on Language in Social Media (LSM 2011)*, pages 30–38, Portland, Oregon. Association for Computational Linguistics.

Faeze Atefinia and Seyed Reza Mirnezami. 2025. Exploring iranian sentiments on the paris agreement: Insights from twitter. *Heliyon*, 11(4):e42716.

Dimitrios Effrosynidis, Georgios Sylaios, and Avi Arampatzis. 2022. Exploring climate change on twitter using seven aspects: Stance, sentiment, aggressiveness, temperature, gender, topics, and disasters.

Pascale Feldkamp, Jan Kostkan, Ea Overgaard, Mia Jacobsen, and Yuri Bizzoni. 2024. Comparing tools for sentiment analysis of Danish literature from hymns to fairy tales: Low-resource language and domain challenges. In *Proceedings of the 14th Workshop on Computational pproaches to Subjectivity, Sentiment, & Social Media nalysis*, pages 186–199, Bangkok, Thailand. Association for Computational Linguistics.

Tamanna Hossain, Robert L. Logan IV, Arjuna Ugarte, Yoshitomo Matsubara, Sean Young, and Sameer Singh. 2020. COVIDLies: Detecting COVID-19 misinformation on social media. In *Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at EMNLP 2020*, Online. Association for Computational Linguistics.

Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. In *Proceedings of the 56th nnual Meeting of the ssociation for Computational Linguistics (Volume 1: Long Papers)*, pages 328–339, Melbourne, Australia. Association for Computational Linguistics.

Maria L. Loureiro and Maria All . 2020. Sensing climate change and energy issues: Sentiment and emotion analysis with social media in the U.K. and Spain. *Energy Policy*, 143(C).

Toni Pano and Rasha Kashef. 2020. A complete vader-based sentiment analysis of bitcoin (btc) tweets during the era of covid-19. *Big Data Cogn. Comput.*, 4:33.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.